Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **NCERT Booster Test Series for NEET-2026_RM(P2)_NBTS-04B** **Time : 180 Min.**

Topics Covered:
Physics: Electromagnetic Waves, Ray Optics & Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor, Experimental Skills
Chemistry: Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids, Amines (Organic Compound containing Nitrogen), The d & f-Block Elements, Coordination Compounds, The p-Block Elements, Biomolecules, Principles Related to Practical Chemistry
Botany: Principles of Inheritance & Variation, Molecular Basis of Inheritance, Microbes in Human Welfare, Organisms and Populations, Ecosystem, Biodiversity and Conservation
Zoology: Human Reproduction, Reproductive Health, Evolution, Human Health & Disease, Biotechnology-Principles and Processes, Biotechnology and Its Applications

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

Vivek Krishna

PHYSICS

1. If F_{pp} , F_{pn} and F_{nn} are the magnitudes of nuclear force between proton-proton, proton-neutron and neutron-neutron respectively, then

(1) $F_{pp} = F_{pn} = F_{nn}$

(2) $F_{pp} < F_{pn} = F_{nn}$

(3) $F_{pp} > F_{pn} > F_{nn}$

(4) $F_{pp} < F_{pn} < F_{nn}$

2. A monochromatic light of frequency 3×10^{14} Hz is produced by a source of power 3×10^{-5} W. The number of photons emitted per second are

(1) 1.5×10^{16}

(2) 1.5×10^{14}

(3) 1.5×10^{15}

(4) 1.5×10^{17}

3. The focal length of the objective of an astronomical telescope is 1.0 m. If the magnifying power of telescope is 20, then the length of telescope for relaxed eye is

(1) 85 cm

(2) 95 cm

(3) 105 cm

(4) 115 cm

4. A p-type semiconductor is obtained by doping:
A. Tetravalent silicon with antimony
B. Tetravalent germanium with arsenic
C. Tetravalent germanium with aluminium
D. Tetravalent silicon with indium
Choose the set of correct statements from the options given below.

(1) A and C only

(2) A, B and C only

(3) C and D only

(4) B, C and D only

5. The dual nature of light is exhibited by

(1) Diffraction and interference

(2) Refraction and interference

(3) Compton effect and diffraction

(4) Photoelectric effect and Compton effect

6. The potential energy of electron in an orbit of radius r in hydrogen atom is (here $k = \frac{1}{4\pi\varepsilon_0}$, e = electronic charge)

(1) $\frac{-ke^2}{r^3}$

(2) $\frac{-ke^2}{r}$

(3) $\frac{-ke^2}{2r^2}$

(4) $\frac{-ke^2}{2r}$

7. Pure Si at 500 K has equal number of electron (n_e) and hole (n_h) concentration of 1.7×10^{15} m^{-3}. On doping with boron, hole concentration n_h increases to 4.7×10^{21} m^{-3}. The doped semiconductor is of

(1) N-type with electron concentration $n_e = 3.5 \times 10^{12}$ m^{-3}

(2) P-type with electron concentration $n_e = 6.14 \times 10^{8}$ m^{-3}

(3) N-type with electron concentration $n_e = 8.2 \times 10^{9}$ m^{-3}

(4) P-type with electron concentration $n_e = 2.89 \times 10^{30}$ m^{-3}

8. The experiment which is used to determine Young's modulus of the material of a given wire, is

(1) Resonance tube experiment

(2) Displacement method

(3) Searle's experiment

(4) Young's double slit experiment

9. In Young's double slit experiment the intensity of light at a point on screen where path difference is λ is I. The intensity of the light at a point where the path difference is $\frac{\lambda}{4}$ is [where λ is wavelength of light]

(1) I

(2) $\frac{3I}{4}$

(3) $\frac{3I}{2}$

(4) $\frac{I}{2}$

10. If M_C is the mass of a nucleus of carbon isotope ($_6C^{13}$), M_P and M_N are the masses of a proton and a neutron respectively, then nuclear binding energy of isotope is

(1) $(-M_C + 6M_P + 7M_N)\,c^2$

(2) $(-M_C + 7M_P + 6M_N)\,c^2$

(3) $(-M_C + 13M_P)\,c^2$

(4) $M_C \cdot c^2$

11. Blue light can eject photo-electron from a photo-sensitive surface while yellow light cannot. Then

(1) Violet light can eject photo-electron

(2) Orange light cannot eject photo-electron

(3) Green light may or may not eject photo-electron

(4) All of the above

12. If potential difference between the plates of capacitor changes at the rate of $\frac{dV}{dt} = 10^4$ V/s and capacitance is 2 μF, then displacement current between the plates will be

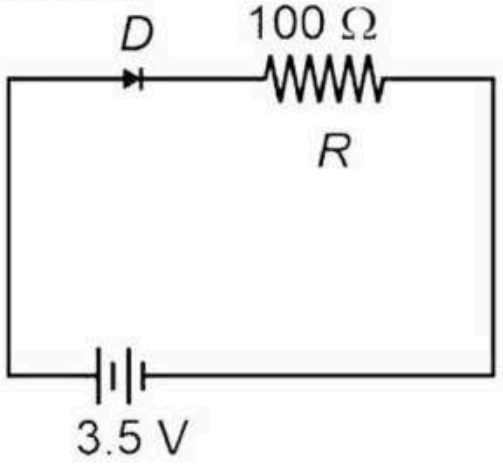
(1) 10 mA

(2) 5 mA

(3) 20 mA

(4) 30 mA

13. In the given figure, a diode D is connected to an external resistance R = 100 Ω and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be



(1) 35 mA

(2) 30 mA

(3) 40 mA

(4) 20 mA

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14. In an electromagnetic wave, the oscillating electric and magnetic field vectors are oriented along

(1) The direction of wave propagation

(2) The same direction but perpendicular to wave propagation

(3) Mutually perpendicular direction which is also perpendicular to direction of propagation

(4) The same direction but at an angle of 30° to wave propagation

15. Avalanche breakdown in a semiconductor diode occurs when

(1) Forward current exceeds a certain value

(2) Reverse bias exceeds a certain value

(3) Forward bias exceeds a certain value

(4) The potential barrier is reduced to zero

16. Refraction of plane wavefront as per Huygen's principle are as shown in the figures. Figure (i) indicates refraction from air to glass and figure (ii) indicates refraction from air to water. In terms of lengths indicated in figures, the refractive index of glass w.r.t. water is

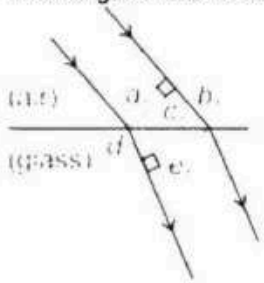


Figure (i)

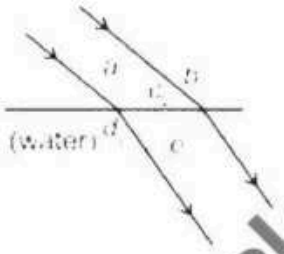


Figure (ii)

(1) $\frac{b_1}{b_2}$

(2) $\frac{d_1}{d_2}$

(3) $\frac{b_1 d_1}{b_2 d_2}$

(4) $\frac{b_1 d_2}{b_2 d_1}$

17. In the Young's double slit experiment, the fringe width, in usual notation is given by

(1) $\beta = \frac{\lambda D}{d}$

(2) $\beta = \frac{\lambda d}{D}$

(3) $\beta = \frac{d}{\lambda D}$

(4) $\beta = \frac{D}{\lambda d}$

18. Match the following w.r.t. spectrum series.

Series	Final orbit(n_f) and initial orbit (n_i) for maximum wavelength
a. Pfund	(i) $n_f = 5, n_i = 6$
b. Brackett	(ii) $n_f = 5, n_i = 4$
c. Paschen	(iii) $n_f = 4, n_i = 5$
	(iv) $n_f = 3, n_i = 4$
	(v) $n_f = 4, n_i = 3$

(1) a(i), b(ii), c(v)

(2) a(i), b(iii), c(iv)

(3) a(ii), b(iii), c(iv)

(4) a(i), b(v), c(iv)

19. Two thin lenses of powers P_1 and P_2 are in contact and coaxial. The focal length of the combination is

(1) $\frac{P_1 P_2}{P_1 + P_2}$

(2) $\frac{P_1 + P_2}{P_1 P_2}$

(3) $P_1 + P_2$

(4) $\frac{1}{P_1 + P_2}$

20. It is given that angular momentum of an electron in a particular orbit of hydrogen like atom of atomic number Z, is L. Assuming Bohr's model and angular momentum quantisation to be true, what will be magnitude of total energy of electron for this particular orbit? (symbols have their usual meanings)

(1) $\frac{mZ^2e^4}{32\pi^2\epsilon_0^2L^2}$

(2) $\frac{mZ^2e^3}{8\pi^2\epsilon_0^2L^2}$

(3) $\frac{mZ^2e^4}{8\pi^2\epsilon_0^2L^2}$

(4) $\frac{m^2Z^2e^4}{8\pi^2\epsilon_0^2L^2}$

21. If 8α and 6β^- particles are emitted in the transformation of $^{238}_{92}U \rightarrow {}^{A}_{Z}Y$, then Z and A respectively will be

(1) 82, 216

(2) 76, 206

(3) 82, 206

(4) 82, 212

22. The magnetic field in a plane electromagnetic wave is given by $B_X = (2 \times 10^{-7}\ \text{T})\sin(ax + bt)$. If wavelength of wave is $\frac{\pi}{50}$ m , then (in appropriate SI units)

(1) $a = 100$

(2) $a = \frac{1}{100}$

(3) $a = 50$

(4) $a = \frac{1}{50}$

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23. Consider the nuclear reaction, $X^{200} \rightarrow A^{110} + B^{90}$
If the binding energy per nucleon for X, A and B is 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, then the amount of the energy released is

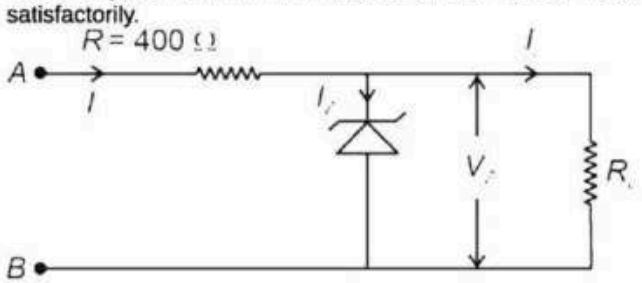
(1) 200 MeV

(2) 160 MeV

(3) 110 MeV

(4) 90 MeV

24. In the given figure, what is the voltage across *AB* needed, to maintain 30 V across the load resistance R_L = 8 kΩ if Zener diode required a minimum current of 25 mA to work satisfactorily.



(1) 41.5 V

(2) 29.5 V

(3) 48.5 V

(4) 50 V

25. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Diffraction is more common in sound as compared to light.
Reason (R) : Usually, wavelength of light is greater than sound.
In light light of the above statements, choose the **correct** answer from the options given below:

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**

(2) Both **A** and **R** are true but **R** is not the correct explanation of **A**

(3) **A** is true but **R** is false

(4) Both **A** and **R** are false

26. Figure of merit of a Galvanometer is defined as

(1) The current required in producing a unit deflection in the scale galvanometer

(2) The deflection produced when a unit current passes through the Galvanometer

(3) The deflection produced when a unit of potential difference is applied between the ends of Galvanometer

(4) The voltage required in producing a unit deflection in the scale of Galvanometer

27. Which of the following waves have the maximum wavelength?

(1) *X* -rays

(2) Radio waves

(3) Ultraviolet waves

(4) Visible rays

28. The diffusion current in *p-n* junction is

(1) From *n* side to *p* side

(2) From *p* side to *n* side

(3) From *n*-type to *p*-side, if junction is forward biased and in opposite direction if it is reversed biased

(4) From *p* side to *n* side if junction is forward biased and in opposite direction if it is reversed biased

29. A convex lens forms an image at a distance 10 cm when a parallel beam of light is incident on it. Find the position(s) of a concave mirror of radius of curvature 5 cm such that the final image is formed at infinity

(1) At a distance of 10 cm from the lens

(2) At a distance of 15 cm from the lens

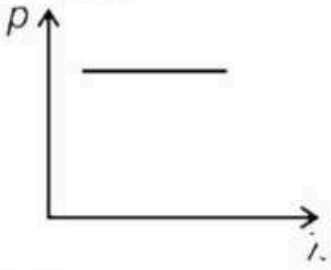
(3) At a distance of 5 cm from the lens

(4) Both (1) & (2)

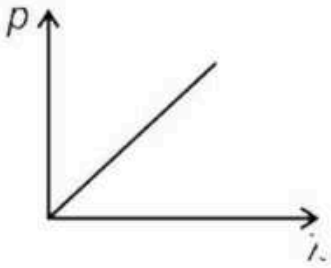
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30. Which of the following figures, represent the variation of particle momentum (p) and the associated de-Broglie wavelength (λ)?

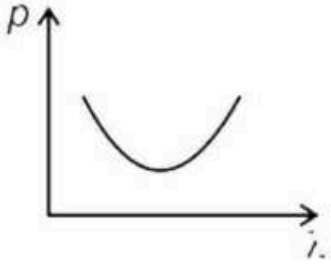
(1)



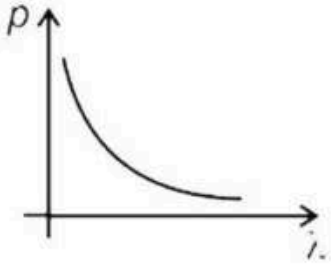
(2)



(3)



(4)



31. Electron in a hydrogen like atom (z = 3) makes transitions from 5th to 4th orbit and from 4th to 3rd orbit. The resulting radiations are incident on a metal plate and eject photo electrons. Stopping potential for photo electrons ejected by shorter wavelength is 3.65 V. What is work function of metal and stopping potential of longer wavelength

(1) 1 eV, 0.75 V

(2) 1 eV, 1 V

(3) 2.30 eV, 1 V

(4) 2.30 eV, 0.45 V

32. A point object is placed at the centre of a glass sphere of radius 4 cm having refractive index 1.5. As seen from outside, the distance of image from the surface is

(1) 4 cm

(2) 2 cm

(3) 1 cm

(4) 6 cm

33. The speed of electromagnetic wave in a medium (whose dielectric constant is 2.25 and relative permeability is 4) is equal to

(1) 0.5×10^8 m/s

(2) 0.25×10^8 m/s

(3) 0.75×10^8 m/s

(4) 1×10^8 m/s

34. A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through analyser B which is oriented such that its pass-axis makes an angle of 30° relative to that of A. The intensity of emergent light is

(1) $\frac{I_0}{4}$

(2) $\frac{3I_0}{8}$

(3) $\frac{3I_0}{4}$

(4) $\frac{I_0}{16}$

35. The angular momentum of an electron in hydrogen atom is $\frac{4h}{2\pi}$. The kinetic energy of this electron is

(1) 1.51 eV

(2) 0.85 eV

(3) 1.70 eV

(4) 3.4 eV

36. Consider the following statements:
a. The density of the nuclear matter is independent of the size of the nucleus.
b. Fusion of light nuclei to form heavier nuclei is possible at room temperature.
c. Electron and positron are particle-antiparticle pair.
The correct statement(s) is/are

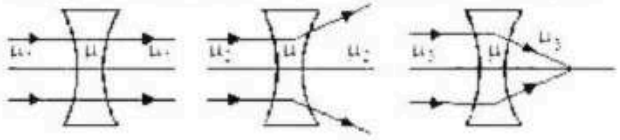
(1) Both a and c

(2) Both a and b

(3) Both b and c

(4) Only b

37. What is the relation between the refractive indices μ_2, μ_1 and μ_3, if the behaviour of light rays is as shown in figure?



(1) $\mu_1 = \mu_2 = \mu_3$

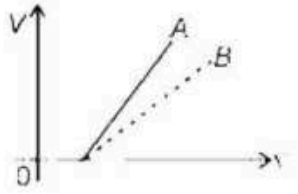
(2) $\mu_1 > \mu_3 > \mu_2$

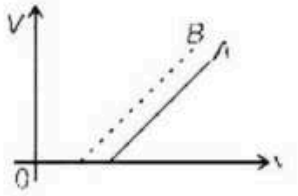
(3) $\mu_3 > \mu_2 > \mu_1$

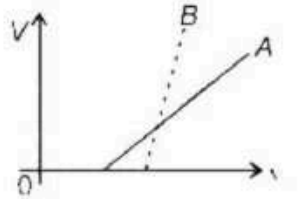
(4) $\mu_3 > \mu_1 > \mu_2$

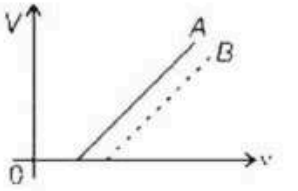
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38. In a photoelectric emission, electrons are ejected from metal *A* and *B* by light of frequency ν. The potential difference *V* required to stop the electrons is measured for various frequencies. If *B* has greater work function than *A*, then graph which illustrates the expected result is

(1) 

(2) 

(3) 

(4) 

39. The focal length of a lens (μ = 1.5) in air is 20 cm. If lens is placed in water (μ = $\frac{4}{3}$) then the focal length of lens in water is

(1) 5 cm

(2) 80 cm

(3) 60 cm

(4) 120 cm

40. The end correction of a resonance column is 1 cm. If the shortest length resonating with tuning fork is 20 cm, the next resonating length should be

(1) 63 cm

(2) 62 cm

(3) 61 cm

(4) 64 cm

41. A nucleus at rest disintegrates into two nuclear parts, which have their speeds in the ratio 8 : 1. Ratio of their nuclear size will be

(1) $2\sqrt{2} : 1$

(2) $1 : 2\sqrt{2}$

(3) $1 : 2$

(4) $1 : 1$

42. **Assertion:** Displacement current is directly proportional to time rate of change of electric flux through given space.

Reason: $I_D = \varepsilon_0 \frac{d\phi_E}{dt}$, (where symbols have their usual meaning.)

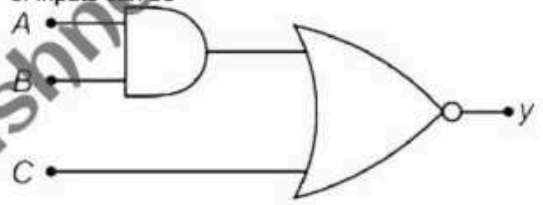
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

43. To get output 0 for the given logic circuit, the correct choice of inputs can be



(1) *A* = 0, *B* = 1, *C* = 0

(2) *A* = 1, *B* = 0, *C* = 0

(3) *A* = 0, *B* = 0, *C* = 0

(4) *A* = 0, *B* = 1, *C* = 1

44. A ray of light, incident on a prism at an angle of incidence 60°, emerges perpendicular to the other face. If the angle of prism is 30°, then the deviation suffered by the ray is

(1) 90°

(2) 60°

(3) 30°

(4) 45°

45. Consider the following statements and choose the correct option.

Statement (A): The impact parameter is the perpendicular distance of the initial velocity vector of the alpha-particles from the centre of nucleus.

Statement (B): For a large impact parameter, the alpha-particle goes nearly undeviated or has a small deflection.

(1) Only Statement (A) is correct

(2) Only Statement (B) is correct

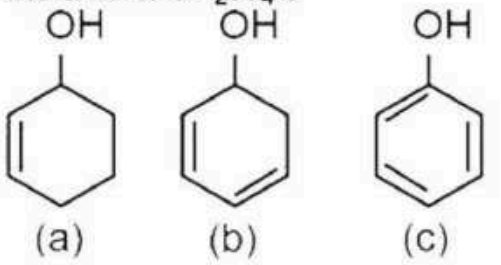
(3) Both Statements (A) and (B) are correct

(4) Both Statements (A) and (B) are incorrect

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CHEMISTRY

46. Decreasing order of rate of dehydration of the following alcohol with conc. H_2SO_4 is



(1) a > b > c

(2) b > a > c

(3) b > c > a

(4) c > a > b

47.



In the above reaction, product formed is

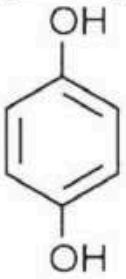
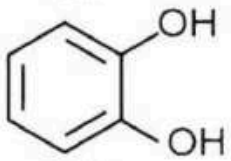
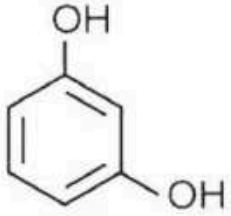
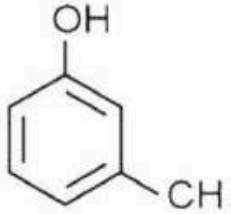
(1) Salicylaldehyde

(2) Aspirin

(3) Benzophenone

(4) Picric acid

48. Match List I with List II

List-I (Name of compound)	List-II (Structure)
a. Resorcinol	(i) 
b. m-Cresol	(ii) 
c. Catechol	(iii) 
d. Quinol	(iv) 

The correct match is

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iv), c(iii), d(i)

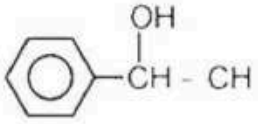
(4) a(ii), b(iii), c(i), d(iv)

49. Alcohol which will react at slowest rate with $ZnCl_2$ and conc. HCl is

(1) CH_3CH_2OH

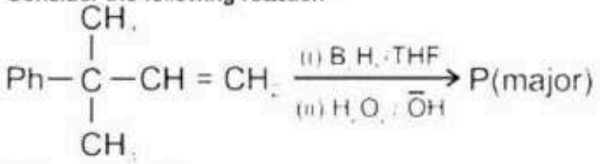
(2) $(CH_3)_3C - OH$

(3) $(CH_3)_2CH - OH$

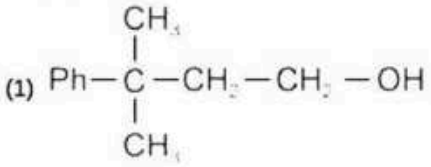
(4) 

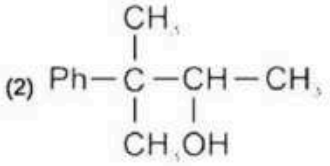
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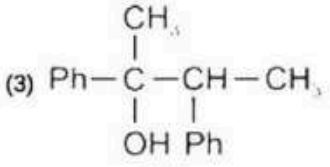
50. Consider the following reaction

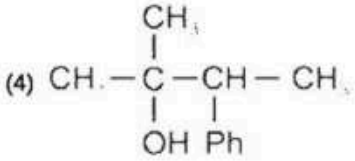


Major product P is

(1) 

(2) 

(3) 

(4) 

51. Williamson synthesis cannot be used for the synthesis of

(1) Diphenyl ether

(2) Dimethyl ether

(3) Ethyl methyl ether

(4) Benzyl phenyl ether

52. Consider the following compounds:
(i) Vanillin
(ii) Acrolein
(iii) Mesityl oxide
The compound(s) in which both –OH and –CHO groups are present is/are

(1) (i) and (ii) only

(2) (i) only

(3) (ii) only

(4) (i), (ii) and (iii)

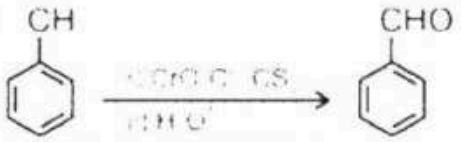
53. Which of the following pairs of compounds cannot be distinguished by iodoform test?

(1) Ethanol and formaldehyde

(2) Ethanol and Pentan-2-one

(3) Acetaldehyde and formaldehyde

(4) Ethanol and propan-1-ol

54. 

The above reaction is known as

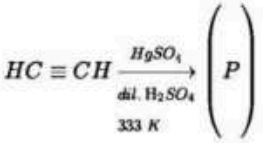
(1) Wolff Kishner reduction

(2) Etard reaction

(3) Gattermann Koch reaction

(4) HVZ reaction

55. Consider the following statements about the product (P) of the given reaction



(a) (P) when heated with dilute NaOH, gives but-2-enal as major product.
(b) It reduces Fehling's reagent.
(c) It gives positive iodoform test.
The correct statements are

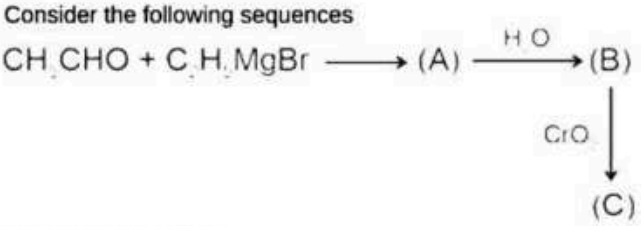
(1) (a) and (b) only

(2) (b) and (c) only

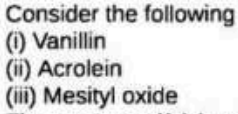
(3) (a) and (c) only

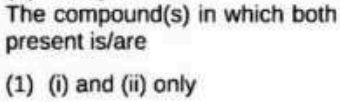
(4) (a), (b) and (c)

56. Consider the following sequences

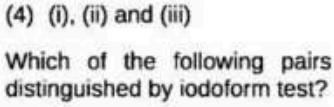


Major product (C) is:

(1) 

(2) 

(3) 

(4) 

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57. Given below are the two statements:
One is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): Aromatic carboxylic acids do not undergo Friedel-Crafts reaction.
Reason (R): Carboxyl group in aromatic carboxylic acid is deactivating in nature and the catalyst aluminium chloride gets bonded to the carboxyl group.
In light of above statements, choose the correct answer from the options given below.

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is correct explanation of A

58. A compound 'X' on reductive ozonolysis gives acetaldehyde and acetone. The compound 'X' is

(1) 2-Methylbut -2-ene

(2) Pent -2-ene

(3) 2-Methylbut-1-ene

(4) But-2-ene

59. The correct decreasing order of basic nature of amines in aqueous medium is–

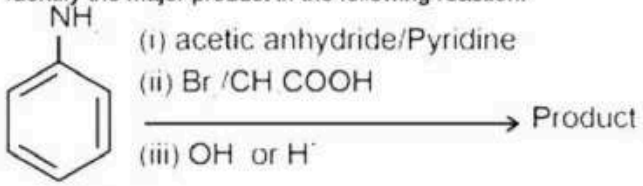
(1) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$

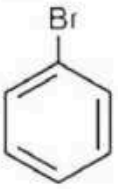
(2) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2$

(3) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$

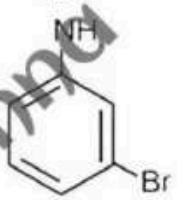
(4) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

60. Identify the major product in the following reaction:



(1)


(2)


(3)


(4)


61. Benzene diazonium chloride on reaction with phenol in alkaline medium forms

(1) p-Hydroxyazobenzene (Blue dye)

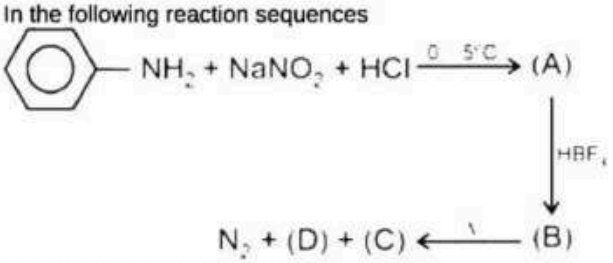
(2) p-Hydroxyazobenzene (Orange dye)

(3) p-Aminoazobenzene (Yellow dye)

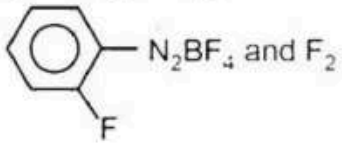
(4) p-Aminoazobenzene (Orange dye)

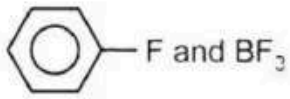
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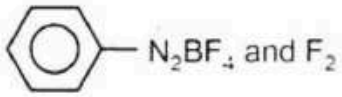
62. In the following reaction sequences

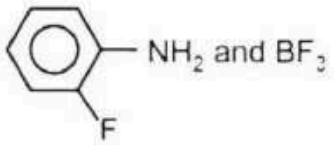


The products (C) and (D) are

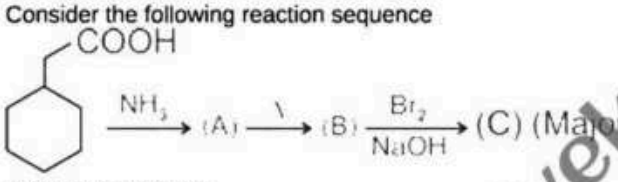
(1) 

(2) 

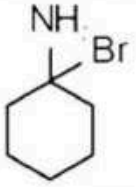
(3) 

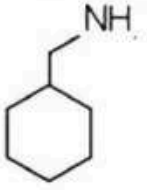
(4) 

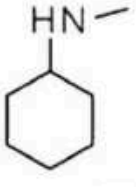
63. Consider the following reaction sequence

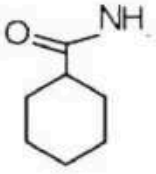


Major product (C) is

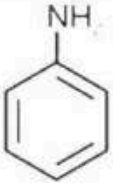
(1) 

(2) 

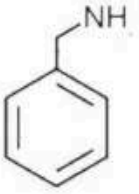
(3) 

(4) 

64. Which among the following compounds cannot be prepared by Gabriel phthalimide synthesis?

(1) 

(2) 

(3) 

(4) 

65. Match the List-I with List-II.

List-I (Carbohydrates)	List-II (Monosaccharide units)
a. Sucrose	(i) β-D-glucose
b. Maltose	(ii) β-D-glucose and β-D-galactose
c. Lactose	(iii) α-D-glucose
d. Cellulose	(iv) α-D-glucose and β-D-fructose

Choose the correct match from the options given below.

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iv), b(iii), c(i), d(ii)

66. Consider the following statements:
(a) Hormones act as intercellular messengers
(b) Hormones are produced by endocrine glands
(c) Thyroxine produced in thyroid gland is an iodinated derivative of amino acid tyrosine
The correct statement(s) is/are

(1) (a) and (b) only

(2) (b) only

(3) (c) only

(4) (a), (b) and (c)

67. Consider the following statements:
Statement-I: Nucleotide contains a sugar molecule and a nitrogenous base only.
Statement-II: Nucleoside contains a sugar molecule, a nitrogenous base and a phosphoric acid molecule.
In the light of above statements, choose the correct option.

(1) Statement-I is correct but Statement-II is incorrect

(2) Statement-I is incorrect but Statement-II is correct

(3) Both the statements-I and II are correct

(4) Both the statements-I and II are incorrect

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68. Choose the incorrect match.

Vitamin	Deficiency disease
(1) Thiamine	Beri-beri
(2) Riboflavin	Rickets
(3) Pyridoxine	Convulsions
(4) Ascorbic acid	Scurvy

(1) (1)

(2) (2)

(3) (3)

(4) (4)

69. Non-essential amino acid among the following is

(1) Valine

(2) Leucine

(3) Isoleucine

(4) Glutamine

70. Which of the following structure(s) of protein get(s) destroyed during its denaturation?
(i) Primary structure
(ii) Secondary structure
(iii) Tertiary structure
The correct option is

(1) (i) and (ii) only

(2) (ii) and (iii) only

(3) (i) only

(4) (i), (ii) and (iii)

71. Consider the following statements:

Statement-I: Pb^{2+} ions react with potassium chromate to form yellow precipitate.
Statement-II: Lead chromate is insoluble in hot sodium hydroxide solution.
In the light of above statements, choose the correct option.

(1) Statement-I is correct and statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I and statement-II are correct

(4) Both the statement-I and statement-II are incorrect

72. Match the List-I with List-II

List-I (Colour of the flame observed by naked eyes)	List-II (Metal ion)
a. Crimson red flame	(i) Cu^{2+}
b. Apple green flame	(ii) Sr^{2+}
c. Brick red flame	(iii) Ba^{2+}
d. Green flame with blue centre	(iv) Ca^{2+}

Choose the correct match from the options given below.

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

73. Given below are two statements
Statement I: Carbonate salts give brisk effervescence on reaction with dilute H_2SO_4.

Statement II: The group reagent for both Al^{3+} and Fe^{3+} is NH_4OH in the presence of NH_4Cl.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

74. Given below are two statements.
Statement I: Both Neptunium and Americium show +7 oxidation states.
Statement II: Both Neptunium and Americium are transuranium elements.
In the light of above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

75. The correct order of ionic radii of Ho^{3+}, La^{3+}, Eu^{3+} and Tm^{3+} is

(1) $La^{3+} < Eu^{3+} < Ho^{3+} < Tm^{3+}$

(2) $Tm^{3+} < Ho^{3+} < Eu^{3+} < La^{3+}$

(3) $Ho^{3+} < Tm^{3+} < Eu^{3+} < La^{3+}$

(4) $Tm^{3+} < Eu^{3+} < La^{3+} < Ho^{3+}$

76. The correct order of oxidising power for following species is

(1) $MnO_4^- > VO_2^+ > Cr_2O_7^{2-}$

(2) $VO_2^+ > Cr_2O_7^{2-} > MnO_4^-$

(3) $MnO_4^- > Cr_2O_7^{2-} > VO_2^+$

(4) $Cr_2O_7^{2-} > VO_2^+ > MnO_4^-$

77. Match the catalyst in List-I with process given in List-II

Catalyst	Process
a. V_2O_5	(i) Oxidation of SO_2 in the manufacture of H_2SO_4 by contact process
b. $PdCl_2$	(ii) Polymerisation of alkynes
c. Fe	(iii) Wacker's process
d. Nickel complexes	(iv) Haber's process

Choose the correct answer

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(i), b(iii), c(iv), d(ii)

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78. Element of most positive value of $E^{\circ}_{M^{3+}/M^{2+}}$ among the following is

(1) Ti

(2) V

(3) Mn

(4) Co

79. Consider the following statements

(a) Spin only magnetic moment value of $[MnBr_4]^{2-}$ is 5.9 BM

(b) trans-$[CrCl_2(ox)_2]^{3-}$ is an optically active species.

(c) Hybridisation of Fe in $[Fe(CN)_6]^{3-}$ is d^2sp^3.

The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

80. Consider the following statements

(a) $EDTA^{4-}$ is a hexadentate ligand.

(b) NO_2^- and SCN^- are ambidentate ligands.

(c) $C_2O_4^{2-}$ and CN^- are chelate ligands.

The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (c) only

81. Correct order of wavelength of visible light absorbed by following complexes is

(1) $[CrCl_6]^{3-} > [Cr(NH_3)_6]^{3+} > [Cr(H_2O)_6]^{3+}$

(2) $[Cr(H_2O)_6]^{3+} > [Cr(NH_3)_6]^{3+} > [CrCl_6]^{3-}$

(3) $[CrCl_6]^{3-} > [Cr(H_2O)_6]^{3+} > [Cr(NH_3)_6]^{3+}$

(4) $[Cr(NH_3)_6]^{3+} > [Cr(H_2O)_6]^{3+} > [CrCl_6]^{3-}$

82. Which among the following complexes is a inner orbital as well as paramagnetic in nature?

(1) $[CoF_6]^{3-}$

(2) $[Fe(CN)_6]^{4-}$

(3) $[Co(NH_3)_6]^{3+}$

(4) $[Mn(CN)_6]^{4-}$

83. How many mole(s) of AgCl will be precipitated if one mole of $[Cr(NH_3)_6]Cl_3$ is added to excess of $AgNO_3$ solution?

(1) 1

(2) 2

(3) 3

(4) 4

84. Consider the following metal carbonyls

a. $[Mn(CO)_6]^+$

b. $[Ni(CO)_4]$

c. $[V(CO)_6]^-$

The correct order of C – O bond length in the metal carbonyls will be

(1) a > b > c

(2) c > b > a

(3) b > a > c

(4) b > c > a

85. Consider the following statements.

(a) Phosphorus trichloride is obtained by the action of thionyl chloride with white phosphorus.

(b) N_2O_5 molecule does not contain N – N bond.

(c) Orthophosphorus acid on heating disproportionates to give orthophosphoric acid and phosphine.

The correct statement(s) is/are

(1) (b) and (c) only

(2) (a), (b) and (c)

(3) (a) and (c) only

(4) (b) only

86. Given below are two statements:

Statement I: For the hydrides of group 16 elements, acidic character increases from H_2O to H_2Te.

Statement II: For the hydrides of group 16 elements, thermal stability decreases from H_2O to H_2Te

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

87. Consider the following statements.

(a) XeF_6 on complete hydrolysis gives XeO_3.

(b) XeO_3 is trigonal planar in shape.

(c) XeO_3 is a colourless explosive solid.

The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (c) only

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88. Consider the following set of reactions.

$Cl_2 + NaOH \xrightarrow[\text{(cold \& dilute)}]{} NaCl + H_2O + A$

$Cl_2 + NaOH \xrightarrow[\text{(hot \& conc.)}]{} NaCl + H_2O + B$

Products A and B respectively are

(1) $NaClO_3$ and $NaClO_4$

(2) NaOCl and $NaClO_4$

(3) $NaClO_3$ and NaOCl

(4) NaOCl and $NaClO_3$

89. Match oxides in List-I with their nature in List-II

List-I	List-II
(a) B_2O_3	(i) Amphoteric
(b) Al_2O_3	(ii) Acidic
(c) In_2O_3	(iii) Neutral
(d) CO	(iv) Basic

Choose the correct answer from the options given below.

(1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(2) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(4) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

90. When dimethyl glyoxime is added to the aqueous solution of nickel chloride in an alkaline medium, it forms a complex having

(1) Black colour

(2) Red colour

(3) Blue colour

(4) Yellow colour

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91. The change in nitrogenous base of template strand of DNA which is responsible for sickle-cell anaemia, is

(1) GAG → GUG

(2) GUG → GAG

(3) CTC → CAC

(4) CAC → CTC

92. In a garden pea, the stem may be tall (T) or dwarf (t) in height. What percentage of the offspring in TT × Tt cross would be expected to be tall?

(1) 75%

(2) 25%

(3) 50%

(4) 100%

93. Find out the **incorrect** statement w.r.t. honey bee.

(1) The females are diploid and males are haploid.

(2) Unfertilised egg develops as a male by means of parthenogenesis.

(3) The females have half the number of chromosomes than that of a male.

(4) Males do not have father, but have grandfather.

94. Cross between which pea plants is considered to be the test cross?

(1) Plant with green pod × Plant with yellow pod

(2) Terminal flowered plant × Terminal flowered plant

(3) Heterozygous round seeded plant × Homozygous round seeded plant

(4) Homozygous violet flowered plant × Heterozygous violet flowered plant

95. Which one of the following traits is expressed only in the presence of identical alleles?

(1) Yellow seed

(2) Inflated pod

(3) Green pod

(4) Green seed

96. Select the **incorrect** statement w.r.t. phenylketonuria.

(1) It is an inborn error of metabolism

(2) The affected individuals lack enzyme phenylalanine hydroxylase

(3) In the affected individual, phenylalanine is converted into tyrosine

(4) Affected individuals experience mental retardation

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97. In *Drosophila*, two recessive genes produce white eye and miniature wings in contrast to their wild type, which results from the dominant alleles. The result from the cross between heterozygous wild type and homozygous recessive one, for both traits are as follows

$w^+ w m^+ m = 66$

$w^+ w m m = 39$

$w w m^+ m = 35$

$w w m m = 60$

In term of cross over units, how far apart are these two genes on the chromosome?

(1) 49 cM

(2) 37 cM

(3) 50 cM

(4) 63 cM

98. Select the option with **correct** set of statements for Punnett square.

(a) Developed by a British geneticist.

(b) Used to calculate the probability of genotypes of offsprings.

(c) Useful for monohybrid crosses only.

(d) It is a mathematical formula which helps to calculate the types of gametes.

(1) Only (a), (b) and (c)

(2) Only (a) and (d)

(3) Only (b) and (d)

(4) Only (a) and (b)

99. Given below is the base sequences of non-coding strand of a transcription unit of DNA

3' T A C A A A G G A A C C A C T 5'

How many amino acids would be coded by the m-RNA formed by this given transcription unit?

(1) 4

(2) 5

(3) 2

(4) 3

100. According to the Chargaff's rule, in a double stranded DNA, if number of adenine = 20%, what is the corresponding percentage of guanine?

(1) 30%

(2) 60%

(3) 35%

(4) 40%

101. During experiment with bacteriophage, Alfred Hershey and Martha Chase (1952) used

(1) P^{32} labelled protein coat and S^{35} labelled DNA

(2) P^{32} labelled DNA and S^{35} labelled protein coat

(3) P^{32} labelled bacteria and S^{35} labelled bacteriophage

(4) P^{32} labelled bacteriophage and S^{35} labelled bacteria

102. Read the following statements and choose the **correct** option.

Assertion (A) : RNA acts as genetic material in few viruses.

Reason (R) : RNA has 2' OH group in its nucleotides.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

103. In eukaryotes, RNA polymerase III transcribes

(1) hnRNA

(2) 28S rRNA

(3) 5.8S rRNA

(4) snRNA

104. In a transcription unit

(1) Promoter is downstream to terminator

(2) Terminator is downstream to structural gene

(3) Promoter sequence codes for RNA

(4) Structural sequence codes for monocistronic RNA in prokaryotes

105. The largest known human gene

(1) Has 3000 bases

(2) Has 30000 base pairs

(3) Is present on chromosome Y

(4) Is dystrophin

106. Which of the following is **incorrect** regarding DNA fingerprinting?

(1) It is based on the use of satellite DNA with high degree of polymorphism called VNTR

(2) Sensitivity of this technique is increased by use of PCR

(3) In this process, characteristic pattern of DNA band differs from individual to individual even for monozygotic twins

(4) It has application in forensic science

107. If 'N' is the population density at time t, then its density at time 't + 1' will be

(1) $N_{t+1} = N_t \times [(B+I) - (D+E)]$

(2) $N_{t+1} = N_t + [(B+I) - (D+E)]$

(3) $N_{t+1} = N_t + [(B+I) + (D+E)]$

(4) $N_{t+1} = N_t \times [(B+D) + (I+E)]$

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108. A population 'X' was reported with the following data.
a. Number of pre-reproductive individuals = 9,875
b. Number of reproductive individuals = 18,960
c. Number of post-reproductive individuals = 11,500
Which of the following conclusions can be withdrawn from the above data?

(1) Such population shows bell-shaped pyramid

(2) Above population will decline with time

(3) This is a stable population

(4) This population will keep on growing always

109. Parasitic mode of life ensures free lodging and meals. In accordance with their life styles, parasites evolved special adaptations such as

(1) Very low reproductive capacity

(2) Loss of unnecessary sense organs

(3) They never show host specificity

(4) They never use two hosts and vectors

110. Which of the given pairs contribute to decrease in population density?

(1) Natality and Immigration

(2) Natality and Mortality

(3) Mortality and Emigration

(4) Immigration and Emigration

111. Which of the following states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eventually eliminated?

(1) Mac Arthur principle

(2) Gause's principle

(3) Tilman's principle

(4) Allen's rule

112. If in a pond there are 80 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 88. The value of birth rate per lotus per year will be

(1) 11

(2) 0.1

(3) 0.9

(4) 1.1

113. Match the **Column I** with **Column II** and select the correct option.

Column I		**Column II**	
a.	Sea anemone and clown fish	(i)	Parasitism
b.	Mycorrhizae	(ii)	Competition
c.	Marine fish and copepods	(iii)	Mutualism
d.	Abingdon tortoise and goat	(iv)	Commensalism

(1) a(iv), b(iii), c(ii), d(i)

(2) a(i), b(iv), c(ii), d(iii)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iii), b(ii), c(iv), d(i)

114. **Assertion (A) :** LAB grow in milk and convert it into curd.
Reason (R) : In our stomach LAB checks the growth of disease causing microbes.
In the light of above statements select the **correct** option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

115. 'Swiss cheese' and 'Roquefort cheese' are ripened by

(1) Bacteria and fungi, respectively

(2) Fungi and bacteria, respectively

(3) Fungi only

(4) Bacteria only

116. Pectinases and proteases are used for

(1) Clarification of bottled juices

(2) Lowering cholesterol levels of blood

(3) Suppressing the immune system of the body

(4) Primary treatment of sewage

117. How many of the following products can be obtained from yeast?
a. Fermented dough of bread
b. Citric acid
c. Alcoholic beverages
d. Blood cholesterol lowering agents
e. Immunosuppressive agent

(1) Five

(2) Four

(3) Three

(4) Two

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118. Measuring Biochemical Oxygen Demand (BOD) is a method used for

(1) Measuring amount of oxygen released by all green plants

(2) Measuring the activity of *Lactobacillus* in producing curd on a commercial scale

(3) Estimating amount of oxygen required by microorganisms to degrade organic matter present in a water sample

(4) Measuring the overall oxygen released by an ecosystem

119. During sewage treatment process, the primary treatment of sewage involves all, **except**

(1) Formation of primary sludge

(2) Use of heterotrophic fungi and bacteria

(3) Filtration of floating debris

(4) Removal of grit

120. Fermented beverages which are produced without distillation are

(1) Whisky and rum

(2) Brandy and beer

(3) Beer and wine

(4) Whisky and brandy

121. A secondary consumer in an ecosystem

(1) Occupies the second trophic level

(2) Is said to be a primary carnivore

(3) Can occupy the first trophic level

(4) Is a herbivorous animal

122. The process in which part of water-soluble substances present in decomposing detritus go down into the soil horizon by percolating water is called

(1) Humification

(2) Fragmentation

(3) Catabolism

(4) Leaching

123. Which of the following statements is **incorrect** regarding decomposition of detritus?

(1) Decomposition rate is slower if detritus is in rich in lignin and chitin

(2) Decomposition rate is faster if detritus is rich in nitrogen and water soluble substances

(3) Warm and moist environment favour decomposition

(4) Cold and dry environment favour decomposition

124. Vertical distribution of different species occupying different levels in ecosystem is called

(1) Stratification

(2) Speciation

(3) Topography

(4) Scarification

125. All of the following are true for ecological pyramids, **except**

(1) It does not take into account the same species belonging to two or more trophic levels.

(2) It assumes a simple food chain.

(3) Saprophytes are not given any place in ecological pyramids.

(4) The base of each pyramid always represents the consumers.

126. For which of the following food chains, pyramid of biomass is inverted?

(1) Grass → Insect → Snake → Frog

(2) Detritus → Earthworms → Sparrow

(3) Phytoplanktons → Zooplanktons → Fish

(4) Tree → Birds → Parasites

127. How many of the following food chains represent upright pyramid of number?
(a) Tree → insects → insect eating birds
(b) Grasses → herbivores → lions
(c) Tree → birds → parasite
(d) Seeds → sparrows → cats
(e) Plants → goats → tigers

(1) Four

(2) Five

(3) Two

(4) Three

128. If in an ecosystem, 3.2 Kcal energy is traped by producers, what will be the amount of energy available for the secondary consumers.

(1) 3200 cal

(2) 3.2 cal

(3) 320 cal

(4) 32 cal

129. `Rivet popper hypothesis' was given by

(1) Alexander von Humboldt

(2) Edward Wilson

(3) Paul Ehrlich

(4) David Tilman

130. According to David Tilman's outdoor plots experiments, which of the following statements is **true** for a stable community?

(1) More diverse community will show less year to year variation in total biomass.

(2) Increased diversity contributes to lesser productivity.

(3) More diverse communities show more year to year variation in total biomass.

(4) Decreased diversity contributes to more productivity.

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131. Select the **incorrect** statement w.r.t. biodiversity richness in tropics.

(1) Tropical environments are less seasonal, relatively more constant and predictable

(2) Warm temperature and high humidity in tropics provide favourable condition throughout the year

(3) Receive less solar energy over the year

(4) They faces less natural disasters

132. Mark true **(T)** or false **(F)** for the following statements and select the **correct** option.
A. Term biodiversity was popularised by a sociobiologist to describe the combined heterogeneity at all levels of biological organisation.
B. More than 50,000 genetically different strains of rice occur in India.
C. Western Ghats have greater amphibian species diversity as compared to Eastern Ghats.
D. India has only 8.1 Per cent of the world's land area, its share of the global speces diversity is only 2.4 per cent.

(1) A-(T), B-(F), C-(F), D-(F)

(2) A-(F), B-(F), C-(T), D-(F)

(3) A-(T), B-(T), C-(T), D-(F)

(4) A-(T), B-(T), C-(F), D-(F)

133. Which one is **not** explained by species-area relationship curve?

(1) On the log scale, relationship between species richness and area is straight line

(2) The slope of the line is much steeper for very small areas

(3) Species richness increases with increasing explored area but only upto a limit

(4) The relation between species richness and area is rectangular hyperbola for variety of taxa

134. Which of the given is an example of organism with recent extinction in Mauritius?

(1) Dodo

(2) Quagga

(3) Thylacine

(4) Steller's sea cow

135. The most important cause driving animals and plants to extinction is

(1) Habitat loss and fragmentation

(2) Alien species invasions

(3) Over-exploitation

(4) Co-extinctions

ZOOLOGY

136. Which of the following structures/cells found in highly coiled tubules in testis, provide nutrition to the germ cells?

(1) Interstitial cells

(2) Sertoli cells

(3) Leydig cells

(4) Immunologically competent cells

137. In human females, which layer of the uterine wall undergoes cyclical changes during reproductive phase?

(1) Endometrium

(2) Myometrium

(3) Perimetrium

(4) Mesometrium

138. Which of the following has 22 + X chromosomes in its nucleus?

(1) Blastomere

(2) Zygote

(3) Ootid

(4) Primary spermatocyte

139. A hormone 'P' responsible for maintaining corpus luteum also acts on 'R' cells in male testes to stimulate the synthesis and secretion of S. Identify 'P', 'R' and 'S' and select the **correct** option.

	P	R	S
(1)	FSH	Leydig cells	Androgens
(2)	LH	Sertoli cells	Estrogen
(3)	FSH	Sertoli cells	Progesterone
(4)	LH	Leydig cells	Androgens

(1) (1)

(2) (2)

(3) (3)

(4) (4)

140. Limbs and external genital organs are formed at which time of pregnancy?

(1) End of 5 weeks of pregnancy

(2) End of 12 weeks of pregnancy

(3) End of second trimester

(4) End of 5 months of pregnancy

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141. Match **List - I** with **List - II.**

List - I	List - II
A. Head	(i) Enzymes
B. Middle piece	(ii) Sperm motility
C. Acrosome	(iii) Energy
D. Tail	(iv) Genetic material

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-I, D-II

(4) A-IV, B-III, C-II, D-I

142. In human females, the signals for parturition induce mild uterine contractions which triggers the release of oxytocin from

(1) Maternal adenohypophysis

(2) Pars intermedia of the foetus

(3) Foetal neurohypophysis

(4) Maternal neurohypophysis

143. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): In humans, reproductive events are identical in both male and female.
Reason (R) : Oogenesis is initiated at puberty due to significant increase in the secretions of GnRH.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

144. In human females, the secretory phase of the uterine cycle
a. Is associated with the increasing levels of progesterone and estrogen
b. Corresponds with the luteal phase of the ovarian cycle
c. Makes the endometrium least receptive to a fertilized embryo
d. Corresponds with the follicular phase of the ovarian cycle
Select the option that represents only the **correct** set.

(1) a and d

(2) b and c

(3) a and b

(4) c and d

145. Which of the following cannot be considered as a reason for increase in population of India?

(1) Decline in infant mortality rate

(2) Decline in maternal mortality rate

(3) Decline in death rate

(4) Decline in number of people in reproducible age

146. Which of the following contraceptives provides protection against STIs?

(1) Diaphragms

(2) Condoms

(3) Vaults

(4) CuT

147. Which of the following statements is **incorrect** w.r.t. amniocentesis?

(1) It involves removal of small sample of amniotic fluid surrounding the foetus.

(2) It is usually done in order to diagnose chromosomal abnormalities and other genetic disorders.

(3) It involves withdrawal of sample of some of the placental tissue.

(4) It is banned in India for sex determination

148. Which of the following is a component of steroidal oral contraceptive pills?

(1) Oxytocin

(2) Relaxin

(3) Progesterone

(4) Cortisol

149. Read the given statements carefully w.r.t India.
Statement A: To tackle with the problem of increasing population growth rate, incentives are given to couples with small families.
Statement B: Statutory raising of marriageable age of the female to 18 years and that of males to 31 years is helpful in checking increasing population growth rate.
Select the **correct** option.

(1) Both statements A and B are correct

(2) Both statement A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

150. An assisted reproductive technique that involves transfer of embryo with 16 celled stage into X of a human female for completion of its further development is termed as Y.
Identify X and Y and select the **correct** option.

(1) X-Fallopian tubes; Y-ZIFT

(2) X-Uterus; Y-IUT

(3) X-Oviducts; Y-GIFT

(4) X-Ampulla of oviducts; Y-ET

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151. Which of the following is/are key concept(s) of Darwin's theory of evolution?
(a) Branching descent
(b) Use and disuse of organ
(c) Natural selection
Choose the option with correct set

(1) Only (b)

(2) Only (b) and (c)

(3) Only (a)

(4) Only (a) and (c)

152. Presence of which of the following will affect the Hardy-Weinberg equilibrium?

(1) Founder effect

(2) Random mating

(3) Stable allelic frequencies

(4) Constant gene pool

153. Koala and Bandicoot show

(1) Adaptive radiation

(2) Convergent evolution

(3) Parallel evolution

(4) Analogy

154. A mice population is in Hardy-Weinberg equilibrium. If in this population, 16% mice are homozygous recessive, then what percentage of the mice population is homozygous dominant?

(1) 16%

(2) 36%

(3) 49%

(4) 64%

155. The ancestor of modern human with brain size more than 900cc was

(1) *Homo habilis*

(2) *Dryopithecus*

(3) *Homo sapiens neanderthalensis*

(4) *Ramapithecus*

156. Type of natural selection in which more individuals acquire peripheral character values at both ends of the distribution curve, is called

(1) Stabilising selection

(2) Directional selection

(3) Disruptive selection

(4) Cyclic selection

157. Bryophytes originated from _____ while the ancestors of psilophyton were _____ .
Choose the correct option to fill in the blanks respectively.

(1) Tracheophyte ancestors, Zosterophyllum

(2) Chlorophyte ancestors, Rhynia-type plants

(3) Tracheophyte ancestors, Lycopods

(4) Psilophyton, Chlorophyte ancestors

158. Consider the following statements.
(a) All existing life forms share similarities to varying degrees and may also share common ancestor.
(b) Geological history of earth closely correlates with the biological history of earth.
(c) *Tyrannosaurus* was the largest herbivore dinosaur.
(d) Evolution of DDT resistant mosquitoes is an example of evolution by anthropogenic action.
How many of the above statements is/are **correct**?

(1) One

(2) Two

(3) Three

(4) Four

159. In a monomeric antibody molecule, the antigen binding site is formed by the

(1) C-terminal of light and N-terminal of heavy chain

(2) N-terminal of light and C-terminal of heavy chain

(3) N-terminals of both light and heavy chains

(4) C-terminals of both light and heavy chains

160. **Assertion (A):** When an antigen enters the body of a person, specific B cells start dividing to produce antibodies.
Reason (R): If the same pathogen enters our body again, memory B cells will produce a primary response.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) is the correct explanation of (A)

(2) Both (A) and (R) are true, (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

161. MALT constitutes about _____ per cent of the lymphoid tissue in human body. Select the **correct** option to fill in the blank.

(1) Twenty

(2) Forty

(3) Fifty

(4) Thirty

162. Match column I with column II w.r.t diseases and their causative agents and choose the **correct** option.

Column I	**Column II**
a. Ringworm	(i) *Wuchereria*
b. Filariasis	(ii) *Ascaris*
c. Common cold	(iii) *Microsporum*
d. Ascariasis	(iv) Rhino virus

(1) a(iv), b(iii), c(i), d(ii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(i), b(iv), c(iii), d(ii)

(4) a(iii), b(i), c(iv), d(ii)

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163. How many diseases given in the box below are vector-borne?

Elephantiasis, Malaria, Common cold, Ascariasis, Chikungunya

Choose the **correct** option.

(1) Two

(2) Four

(3) Three

(4) Five

164. Marijuana and charas are products of

(1) *Papaver somniferum*

(2) *Cannabis sativa*

(3) *Atropa belladona*

(4) *Datura*

165. **Assertion (A) :** Cancer cells can spread to distant sites from original location through blood.
Reason (R) : Cancer cells possess the property called contact inhibition.
In the light of above statements, select the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of the (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

166. Read the following statements.
(a) Plasmid is chromosomal linear autonomously replicating DNA.
(b) *rop* codes for the proteins involved in the replication of the plasmid.
(c) Maintenance of introduced DNA in host and transfer of this DNA to its progeny is essential during genetic modification of an organism.
How many of the above statements is/are **incorrect**?

(1) Zero

(2) One

(3) Two

(4) Three

167. If one wants to generate a high copy number of a desired DNA segment, then he/she should take a vector which

(1) Has many restriction sites for a single restriction enzyme

(2) Is bigger in size

(3) Has selectable markers controlling the copy number

(4) Has the *ori* site to support high copy number

168. If a circular dsDNA possesses three restriction sites for *Eco*RI and was subjected to digestion with the same restriction enzyme, then after complete digestion, the mixture will contain

(1) 3 fragments

(2) 4 fragments

(3) 5 fragments

(4) 2 fragments

169. During gel electrophoresis, the separated DNA fragments can be visualised only after staining with 'X' followed by exposure to 'Y'. Select the option that **correctly** identifies 'X' and 'Y' respectively.

(1) Ethidium bromide, X-rays

(2) Methylene blue, Visible light

(3) Methylene blue, UV rays

(4) Ethidium bromide, UV radiation

170. Select the **incorrect** option w.r.t. the steps involved in the post-biosynthetic stage of RDT.

(1) Quality control testing

(2) Formulation of products with suitable preservatives

(3) Separation and purification of the desired product

(4) Culturing the host cells with rDNA in a medium at large scale

171. The construction of the first recombinant DNA by Stanley Cohen and Herbert Boyer was done in the year

(1) 1980

(2) 1971

(3) 1982

(4) 1972

172. Sampling ports in a bioreactor are meant for

(1) Temperature control system

(2) pH control system

(3) Introduction of raw material

(4) Withdrawing small volumes of culture periodically

173. At what temperature, the heat shock treatment is given that enables the bacteria to take up the rDNA during bacterial transformation?

(1) 52°C

(2) 62°C

(3) 42°C

(4) 72°C

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174. "The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services."
The above definition of biotechnology was given by

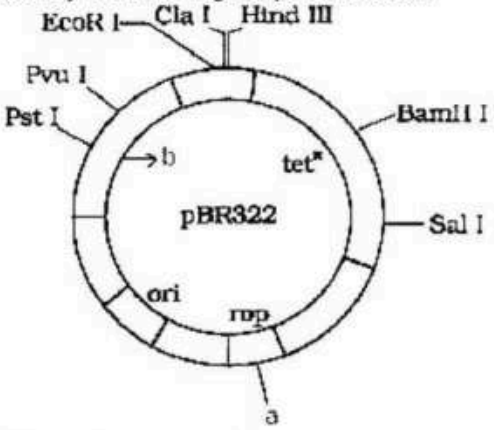
(1) EFB

(2) GEAC

(3) NDRI

(4) CDRI

175. Identify 'a' and 'b' in given plasmid vector.



Choose the correct option.

(1) a = *Pvu*-II, b = amp^R

(2) a = *Hind* II, b = tet^R

(3) a = *Sal* I, b = *ori*

(4) a = *Hind* III, b = tet^R

176. In mature form of human insulin, the two polypeptide chains are linked to each other by

(1) Phosphodiester bonds

(2) Disulphide bonds

(3) Glycosidic bonds

(4) Hydrogen bonds

177. Which of the following is true w.r.t. RNAi?

(1) Involves silencing of the target mRNA by preventing its translation.

(2) *M. incognita* was used as a vector to get rid of *Agrobacterium*.

(3) DNA is introduced in such a manner that only sense RNA is produced in the host cells.

(4) Takes place only in prokaryotic organisms as a method of cellular defense

178. Select the **incorrect** option w.r.t. tissue culture.

(1) Regeneration of whole plant from any part of plant under sterile conditions in a special nutrient media comes under this technology.

(2) Using this technique, a large number of plants cannot be formed from the same plant in a very short duration.

(3) The plants regenerated is genetically identical to the original plant from which the explant is taken.

(4) Banana has been produced on commercial scale using this method.

179. Read the following statements.
Statement A: Genetically modified crop is used to maximise the use of fertilizers and chemicals so that their harmful effects on the environment are reduced.
Statement B: Genetic modification has been used to create tailor-made plants to supply alternative resources to industries , in the form of starches, fuel and pharmaceuticals.
Select the **correct** option.

(1) Only statement B is correct

(2) Only statement A is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

180. How many of the statement(s) given below are true?
(a) In 1999, the first transgenic cow, Rosie, produced human protein-enriched milk.
(b) ELISA is based on the principle of antigen-antibody interaction.
(c) In 1983, Eli Lilly an American company prepared two DNA sequences corresponding to A and B chains of human insulin.
(d) In 1992, the first clinical gene therapy was given to a 4-year-old girl with adenosine deaminase (ADA) deficiency.
Select the correct option.

(1) Two

(2) Three

(3) One

(4) Four

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